

Analysis of the App

Privacy Analysis of Augmented Reality Glasses

1 Set Up MobSF

1. Make sure to have Docker installed on your system. The instructions for downloading on an Ubuntu machine can be found below in the section 'Install Docker'.
2. Get the apk file from the phone, the instructions can be found below in the section 'Get APK'.
3. Open a terminal window on your computer
4. Download the latest MobSF docker image with the command:

```
sudo docker pull opensecurity/mobile-security-framework-mobsf:latest
```

2 Static Analysis

1. Start MobSF for static analysis in Docker using the command which starts the MobSF server and maps it to port 8000 on your computer:

```
sudo docker run -it --rm \
  -p 8000:8000 \
  opensecurity/mobile-security-framework-mobsf:latest
```

Wait till you see in the output in the terminal indicating that the server has started and is listening on a specific port (8000), it should look like:

```
[2024-11-07 17:54:02 +0000] [1] [INFO] Starting gunicorn 23.0.0
[2024-11-07 17:54:02 +0000] [1] [INFO] Listening at: http://0.0.0.0:8000 (1)
[2024-11-07 17:54:02 +0000] [1] [INFO] Using worker: gthread
[2024-11-07 17:54:02 +0000] [125] [INFO] Booting worker with pid: 125
```

2. Open a browser and access the MobSF web interface by entering

```
http://127.0.0.1:8000
```

3. The credentials for MobSF, by default, is:

```
username: mobsf
password: mobsf
```

4. Drag and drop your apk file (base.apk) onto the page and wait for MobSF to create a static analysis. This can take a couple minutes, depending on your app it may take longer.
5. To stop press Ctrl+C in the terminal and close the tab.

3 Static + Dynamic Analysis

1. Set up and start an emulator. Note that there are certain versions that are not supported by MobSF. The instructions to set up an emulator using 'Genymotion Android VM' can be found below in the section 'Emulator'. More information on other supported versions and emulators can be found in the link below, though if you are using the Meta View App, do not use 'Android Studio Emulator' as there are no versions that is both supported by MobSF and the app:

https://mobsf.github.io/docs/#/dynamic_analyzer

2. For Linux based systems, like myself (Ubuntu), make sure that your Docker version is at least 20.04. To check your docker version, use

3. Find the adb device identifier. If you are using Genymotion Android VM, you can also find the identifier on the top of your emulated device. If the emulator is running, you can use the command:

```
adb devices
```

The output should look like:

```
List of devices attached
127.0.0.1:6555  device
emulator-5554  device
```

4. Start MobSF for dynamic analysis in Docker

- for Linux based systems use the following command and replace emulator-5554 with the device identifier found:

```
docker run -it --rm \
-p 8000:8000 \
-p 1337:1337 \
-e MOBSF_ANALYZER_IDENTIFIER=emulator-5554 \
--add-host=host.docker.internal:host-gateway \
opensecurity/mobile-security-framework-mobsf:latest
```

- for other systems, use the following command and replace emulator-5554 with the device identifier found:

```
docker run -it --rm \
-p 8000:8000 \
-p 1337:1337 \
-e MOBSF_ANALYZER_IDENTIFIER=emulator-5554 \
opensecurity/mobile-security-framework-mobsf:latest
```

This starts MobSF for dynamic analysis in Docker which starts the MobSF server and maps it to port 8000 on your computer. If you receive errors, look below in the section 'Errors encountered' to see if I also had that problem and how I solved it. If not, wait till you see in the output in the terminal indicating that the server has started and is listening on a specific port (8000), it should look like:

```
[2024-11-09 23:27:51 +0000] [1] [INFO] Starting unicorn 23.0.0
[2024-11-09 23:27:51 +0000] [1] [INFO] Listening at: http://0.0.0.0:8000 (1)
[2024-11-09 23:27:51 +0000] [1] [INFO] Using worker: gthread
[2024-11-09 23:27:51 +0000] [125] [INFO] Booting worker with pid: 125
```

5. Open a browser and access the MobSF web interface by entering

```
http://127.0.0.1:8000
```

6. The credentials for MobSF, by default, is:

```
username: mobsf
password: mobsf
```

7. Drag and drop your apk file (base.apk) onto the page and wait for MobSF to create a static analysis. This can take a couple minutes, depending on your app it may take longer.

8. To run a dynamic analysis, press 'start dynamic analysis' on the left bar, which should lead you to a different page.

- If you see errors, look below in the section 'Errors encountered' to see if I also had that problem and how I solved it.
- Otherwise, you should see your app and press the green button that says, 'start dynamic analysis' to interact with your app and do an analysis.

9. To stop, close the tab and press Ctrl+C in the terminal.

4 Install Docker

There are some prerequisites and limitations before installing Docker:

1. Firewall limitations: When using `ufw` or `firewalld`, exposing container ports with Docker bypasses these firewall rules. Docker only supports `iptables-nft` and `iptables-legacy` but not rules created with `nft`. Ensure any firewall rules are created with `iptables` or `ip6tables` and added to the `DOCKER-USER` chain.
2. OS Requirements: Docker Engine requires a 64-bit version of Ubuntu Noble 24.04 (LTS), Jammy 22.04 (LTS), or Focal 20.04 (LTS), and is compatible with `x86_64`, `armhf`, `arm64`, `s390x`, and `ppc64le` architectures.

I installed docker using the apt repository. Steps to install docker using the apt repository:

1. Uninstall old versions or conflicting packages of Docker if there is any using the command though I did not have any:

```
for pkg in docker.io docker-doc docker-compose docker-compose-v2
podman-docker containerd runc; do sudo apt-get remove $pkg; done
```

2. Set up Docker's apt repository first by adding Dock's official GPG key:

```
sudo apt-get update
sudo apt-get install ca-certificates curl
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o
/etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc
```

3. Add the repository to Apt sources:

```
echo
"deb [arch=$(dpkg --print-architecture)
signed-by=/etc/apt/keyrings/docker.asc]
https://download.docker.com/linux/ubuntu
$(. /etc/os-release && echo "$VERSION_CODENAME") stable" |
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
sudo apt-get update
```

4. Install the latest Docker packages:

```
sudo apt-get install docker-ce docker-ce-cli containerd.io
docker-buildx-plugin docker-compose-plugin
```

To verify that the installation was successful run:

```
sudo docker run hello-world
```

You should see something like:

```
Hello from Docker!
This message shows that your installation appears to be working correctly.
...
```

The following link shows more detail as well as other methods for installation for Ubuntu:

<https://docs.docker.com/engine/install/ubuntu/>

5 Get APK

1. Connect your phone to your computer via USB and open a terminal window.
2. First you need to know the package name of the app. The name of the Meta View app is called:

```
com.facebook.stella
```

If you want to look for another app, use the following command in the terminal that outputs the list of package names:

```
adb shell pm list packages
```

You can try to filter the names if you have an idea what may be related to (also the company), but know that the name of the package can be different from the name of the app (as seen for the Meta View app):

```
adb shell pm list packages | grep -i "related_name_1 \| related_name_2"
```

Or you can filter by the installation date, using:

```
adb shell pm list packages -3
```

3. After finding out the name, you are able to get the full path name of the apk file. The following is an example for the Meta View app, for a different app, change the name of the package to the one you want:

```
adb shell pm path com.facebook.stella
```

The output should look something like this:

```
package: /data/app/~~RN3FgfL6QR9KkBav3MNe4g==/com.facebook.stella-i-Y3XkQ  
Ks4B2DUrcEwHUBQ==/base.apk
```

4. With the path name, pull the apk file from the phone to your current working directory, using:

```
abd pull /data/app/~~RN3FgfL6QR9KkBav3MNe4g==/com.facebook.stella-i-Y3XkQ  
Ks4B2DUrcEwHUBQ==/base.apk
```

To pull the apk file to a specific directory, use:

```
abd pull path_you_just_got path_you'd_like_to_pull_to
```

You should see a file with the name in your selected directory:

```
base.apk
```

6 Set Up Emulator

MobSF supports Genymotion Android VM with x86, x86_64 architectures, Android versions 4.1 to 11.0, and upto API 30. The Meta View App requires at least Android version 10.0. Most Android phones use ARM architecture, so an ARM translator is needed. I found a translator for Android version 11.0.

7 Errors Encountered